NICHOLAS YAP

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SKILLS

Technical Skills: SolidWorks, Onshape CAD, Bluebeam, SAP GUI

Instrumentation: Oscilloscope, Function Generator, Digital Multimeter, Power Supply

Programming: C, C++, C#, Python, MATLAB, WinForms, PLC Ladder Logic

Embedded Systems: MSP430FR5739 Firmware Development, Code Composer Studio, Soldering

PROJECTS

Self-Balancing Wheeled Bipedal Robot: Prototype Grass – *Paired Project* **November 2025 – February 2026**

- Built a dual-PID self-balancing control system in collaboration with a partner to balance an elevated robot on two motorized wheels through accelerometer, gyroscope, and encoder input.
- Developed a C# WinForms application to interface with an ESP-WROOM-32 over BLE communication, allowing for wireless real-time PC control of the ESP through a client-server relationship.
- Designed and built a modular, multi-component chassis to integrate electrical systems, ensure precise sensor alignment, mitigate signal interference, and incorporate protective features to enhance safety and manufacturability.

Soft-Robotics Octopus Claw Prototype – *Prototype Development/Capstone* **October 2025 – April 2026**

- Researched and developed prototypes of a TPU-structured flexible manipulator inspired by logarithmic spirals in nature to allow for complex 3D movement and adaptable grasping capabilities.

Stepper Motor 2-Axis Gantry – *Paired Project* **September 2025 – December 2025**

- Soldered a predesigned PCB to implement a MSP430FR5739 microcontroller with motor drivers and encoder inputs.
- Programmed firmware in Code Composer Studio to control stepper motors for G-code–based gantry operation via UART from a C# WinForms application, enabling precise image drawing through custom G-code paths.

ENGINEERING EXPERIENCE

The Dow Chemical Company **February 2024 – August 2024**

Maintenance/Reliability Engineering Co-op Fort Saskatchewan, AB

- Reprogrammed an industrial oven PID controller and optimized equipment cleaning procedures, improving system reliability and enhancing maintenance efficiency and operator safety.
- Designed SolidWorks flange covers for shut-down pressure equipment and coordinated Management of Change (MOC) for facility upgrades, ensuring safe implementation, regulatory compliance, and accurate technical documentation.
- Collaborated with vendors, work coordinators, and operators to streamline equipment bill of materials, inventory management, and procurement of critical equipment.

The Dow Chemical Company **September 2023 – February 2024**

Fixed Asset Integrity Engineering Co-op Fort Saskatchewan, AB

- Collaborated with multiple plant operations teams to author and update Corrosion Control Documents and Integrity Operating Windows, improving reliability across all site operations.
- Revised ABSA regulatory documentation to ensure regulatory compliance and maintain the site's permission to operate.

UBC Uncrewed Aircraft Systems **September 2022 – Present**

Senior Aircraft and Administrative Member Vancouver, BC

- Designed airfoil frames, landing gear, and electrical equipment mounts for a VTOL reconnaissance aircraft.
- Collaborated with an interdisciplinary team to develop comprehensive quadcopter and VTOL CAD assemblies in Onshape, creating functional autonomous drones for competition.

EDUCATION

University of British Columbia **May 2026**

Bachelor of Applied Science in Mechanical Engineering, Mechatronics Specialization Vancouver, BC

CGPA: 87.6% – Dean's List | Graduated with Distinction